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COMPUTER NETWORKS

CY245AT

Experiential Learning Report

On

**“Modeling Disaster Spread and Social
Signal Diffusion with Graph
Intelligence”**

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Chapter 1: Introduction

- Natural disasters are increasing in frequency and intensity, requiring responsive, real-time systems for communication and coordination.
- Traditional static systems are too slow and fragile for dynamic disaster scenarios.
- Our project, “Analysis of Information Diffusion in Online Social Networks Using Graph Models,” integrates Computer Networks to build a scalable and resilient disaster dashboard.
- The system uses:
 - REST APIs for data exchange between frontend and backend.
 - WebSocket simulations for low-latency, bidirectional communication.
 - MQTT protocol for efficient IoT sensor data transmission.
- Key problems addressed by the CN component:
 - Reduces human dependency through automated updates from social media and sensors.
 - Simulates real-time alert propagation using WebSocket models.
 - Demonstrates resilience through distributed modular design.
 - Visualizes packet routing and information flow across nodes.
- The dashboard architecture applies core CN principles such as:
 - Protocol layering (MQTT, HTTP, WebSockets),
 - Load balancing (via Nginx or similar proxy),
 - Real-time streaming and edge processing,
 - Fault tolerance in communication layers.
- Overall, the CN component bridges the gap between theoretical models and practical deployment in disaster management systems.

Chapter 2: Design and Implementation

2.1 Network Protocols and Data Flow Architecture

The system employs MQTT (Message Queuing Telemetry Transport) for low-overhead data transmission from IoT sensor nodes to the cloud server. MQTT operates on top of TCP and follows a publish-subscribe model, where sensors publish data and the dashboard subscribes to updates.

HTTP/REST APIs are used for standard web-based data access from social media APIs and third-party feeds. This integration ensures that the backend aggregates information from Twitter, - Reddit, and news portals via GET and POST requests.

2.2 WebSockets for Real-Time Updates

To maintain a persistent, bidirectional communication channel between frontend and backend, WebSocket protocol is integrated. Unlike REST APIs that require polling, WebSockets enable instant push notifications to update alerts, sensor readings, and analytics every 10 seconds. This design drastically reduces server load and improves response time.

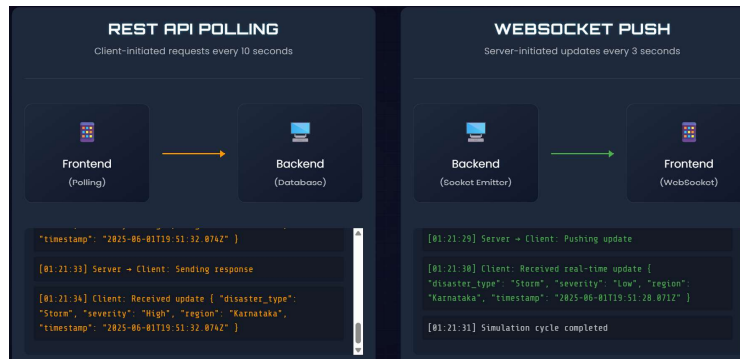


Fig 1. Rest API vs Web socket comparison

2.3 Load Balancing and Scalability

Nginx is used as a reverse proxy and load balancer to ensure high availability and even distribution of incoming requests. It directs WebSocket traffic, RESTful endpoints, and static files to appropriate services. This design supports horizontal scaling in the event of high disaster traffic scenarios, especially during national emergencies.

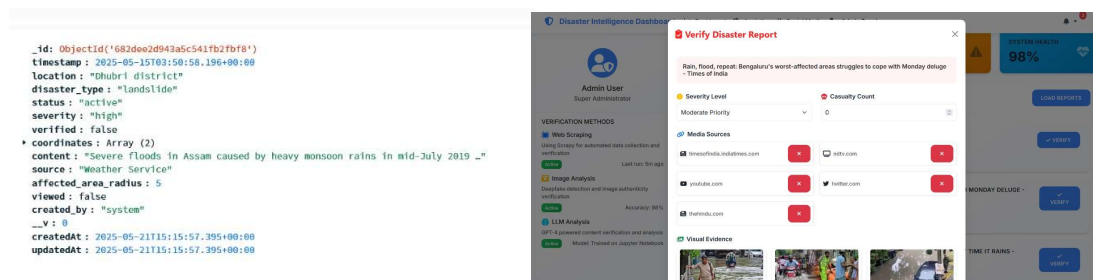


Fig 2.1 & 2.2, mongo DB for scalability (vertical scaling) and result

2.4 Network Security Measures

Secure Sockets Layer (SSL) is implemented to encrypt data transmission over HTTP and WebSocket connections. Authentication headers and JSON Web Tokens (JWT) ensure that only authorized users (e.g., admins, responders) can send or verify alerts. Rate limiting is enforced at the Nginx layer to mitigate DDoS attacks.

2.5 Edge Computing in Network Architecture

Edge nodes are deployed near disaster-prone zones to perform preliminary filtering and aggregation of IoT data. These nodes reduce upstream bandwidth usage by sending only relevant or anomaly-detected events. This minimizes latency and helps prioritize urgent updates.

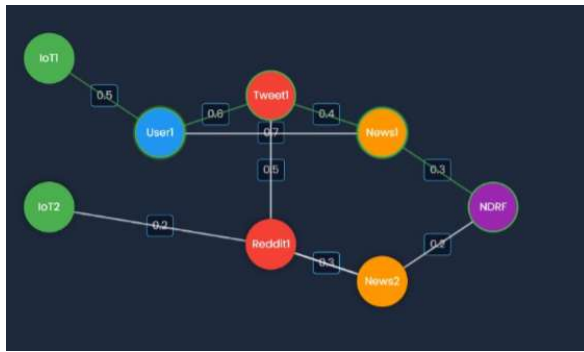


Fig 3. Node-Edge design for data traversal

Chapter 3: Conclusion

- The integration of computer networks in the Disaster Information Dashboard is critical for the system's effectiveness. From real-time communication protocols like WebSockets and MQTT to load balancing and secure data exchange, the network layer forms the backbone of disaster data processing. Future enhancements may include 5G-based data relays, AI-based network traffic prediction, and multi-region cloud deployments to improve latency and uptime.
- The use of RESTful APIs allowed seamless interaction between backend data (hosted in MongoDB) and the frontend dashboard, ensuring that disaster data remains synchronized.
- Simulated WebSockets provided a realistic demonstration of low-latency updates and persistent server-client communication ideal for critical notifications.
- MQTT protocol usage for IoT data simulation helped demonstrate lightweight and efficient data transmission suitable for resource-constrained environments.
- Implementation of edge processing principles reduced network load by simulating on-device filtering and aggregation before transmission.
- Resilience techniques such as fallback fetch strategies, retry mechanisms, and modular architecture ensured the system remained operational during partial failures.
- Simulations included visual feedback loops representing data traveling through network layers, reinforcing learning of OSI layers, routing logic, and latency bottlenecks.

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